

Oral sildenafil citrate (viagra) for erectile dysfunction: a systematic review and meta-analysis of harms.



OBJECTIVES: To summarize and compare evidence on harms in sildenafil- and placebo-treated men with erectile dysfunction (ED) in a systematic review and meta-analysis. **METHODS:** Randomized placebo-controlled trials (RCTs) were identified using an electronic search in MEDLINE, EMBASE, PsycINFO, SCOPUS, and Cochrane CENTRAL. The rates of any adverse events (AEs), most commonly reported AEs, withdrawals because of adverse events, and serious adverse events were ascertained and compared between sildenafil and placebo groups. The results of men with ED were stratified by clinical condition(s). Statistical heterogeneity was explored. Meta-analyses based on random-effects model were also performed. **RESULTS:** A total of 49 RCTs were included. Sildenafil-treated men had a higher risk for all-cause AEs (RR = 1.56, 95% CI: 1.38, 1.76), headache, flushing, dyspepsia, and visual disturbances compared with placebo-treated men. The magnitude of excess risk was greater in fixed- than in flexible-dose trials. The rates of serious adverse events and withdrawals because of adverse events did not differ in sildenafil vs placebo groups. A higher dose of sildenafil corresponded to a greater risk of AEs. The increased risk of harms was observed within and across clinically defined specific groups of patients. **CONCLUSIONS:** There was a lack of RCTs reporting long-term (>6 months) harms data. In short-term trials, men with ED randomized to sildenafil had an increased risk of all-cause any AEs, headache, flushing, dyspepsia, and visual disturbances. The exploration of different modes of dose optimization of sildenafil may be warranted.